An Exact Periodic Continued Fraction Representation Linking Dimensionless Physical Coupling Constants

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(Dated: August 26, 2026)

This research note reports an anomalous mathematical alignment discovered within the regular continued fraction expansions of the reciprocal fine-structure constant (α^{-1}) and the electroweak coupling constant at the Z-boson scale. We demonstrate that both fundamental physical parameters can be modeled using an identical 9-period infinite tail, $[1; 3, 1, 1, 18, 1, 7, 1, 3]$, combined with distinct integer prefixes. The resulting algebraic closed-form solutions yield numerical values that fall precisely within the current 2022 CODATA recommended experimental standard uncertainties. A direct Möbius transformation connecting both scales is derived under the modular group $\mathrm{GL}_2(\mathbb{Z})$, offering a structural bridge for further evaluation in digital physics and experimental number theory.

I. INTRODUCTION

The regular continued fraction representation of a real number provides a unique topological signature that is highly sensitive to its algebraic properties. While standard decimal expansions mask structural relationships, periodic continued fractions identify quadratic irrationalities uniquely, as established by Lagrange's Theorem.

In quantum field theory, dimensionless coupling constants like the fine-structure constant (α^{-1}) are treated as running empirical parameters measured at specific energy scales Q^2. In this note, we demonstrate a remarkably tight numerical bridge connecting two distinct physical energy states using a shared, infinite periodic integer sequence.

II. THE SHARED PERIODIC TAIL

Let us define the core sequence of our investigation as an infinite, self-similar periodic continued fraction tail denoted by x:

$$x = \overline{[1; 3, 1, 1, 18, 1, 7, 1, 3]} \tag{1}$$

By expanding this nested fraction algebraically, x satisfies the self-referential equation:

$$x = 1 + \cfrac{1}{3 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{18 + \cfrac{1}{1 + \cfrac{1}{7 + \cfrac{1}{1 + \cfrac{1}{3 + \frac{1}{x}}}}}}}}} \tag{2}$$

Collapsing this matrix cascade yields the fractional linear equation:

$$x = \frac{6124x + 1575}{4767x + 1226} \tag{3}$$

which simplifies to the quadratic form:

$$4767x^2 - 4898x - 1575 = 0 \tag{4}$$

The unique positive root defining this geometric attractor is exactly:

$$x = \frac{2449 + \sqrt{13505626}}{4767} \approx 1.284665437 \tag{5}$$

III. ALGEBRAIC CONSTANTS AND CODATA ALIGNMENT

By appending specific finite integer prefixes to this exact tail x, we map out two distinct algebraic numbers (A and B) representing theoretical values for the constants under evaluation.

A. Constant A: The Fine-Structure Constant (α^{-1})

Applying the prefix $[137; 27]$ yields:

$$A = [137; 27, x] = 137 + \frac{1}{27 + \frac{1}{x}} = \frac{3700x + 137}{27x + 1} \tag{6}$$

Substituting the exact root for x and rationalizing the denominator gives the closed-form expression:

$$A = \frac{69284635}{505581} - \frac{\sqrt{13505626}}{1011162} \tag{7}$$

Evaluating this numerically yields:

$$A = 137.0359991770497... \tag{8}$$

The latest internationally accepted physical reference value (CODATA 2022 recommendation) lists α^{-1} as:

$$\alpha^{-1}_{\mathrm{CODATA}} = 137.035999177(21) \tag{9}$$

Remarkably, the algebraic evaluation of A sits precisely at the center of this experimental 1-sigma standard uncertainty interval.

B. Constant B: Electroweak Coupling Scale

Applying the prefix $[128; 1, 17]$ to the same tail yields:

$$B = [128; 1, 17, x] = \frac{2321x + 129}{18x + 1} \tag{10}$$

Which resolves exactly to:

$$B = \frac{53814700}{417369} + \frac{\sqrt{13505626}}{417369} \approx 128.946747361 \tag{11}$$

This values mirrors the running fine-structure constant at high energy levels, specifically aligning with the effective coupling value observed at the Z-boson mass scale ($M_Z \approx 91.19$ GeV), conventionally approximated in literature near ~ 128.94.

IV. TOPOLOGICAL CONNECTION VIA MÖBIUS TRANSFORMATION

Because Constants A and B share an identical infinite continued fraction tail, they belong to the same orbit under the modular group $\mathrm{GL}_2(\mathbb{Z})$. By eliminating the shared variable x, we derive a direct, integer-coefficient Möbius transformation that maps the value of A directly onto B:

$$B = \frac{-1162A + 159323}{-9A + 1234} \tag{12}$$

This fractional linear relation proves that the scaling distance between these two critical coupling states is governed by a rigid, non-linear algebraic trajectory.

V. CONCLUSION AND OPEN QUESTIONS

Whether this tight mathematical coherence reflects a genuine fractal or discrete structure of the vacuum energy scale (renormalization group path) or represents an extraordinary numerical coincidence remains an open question. However, the fact that an identical 9-period tail satisfies both highly sensitive parameters within modern experimental thresholds warrants formal archiving. This exact relation provides a concrete, testable framework for upcoming precision measurement cycles in sub-atomic physics.

ACKNOWLEDGMENTS

The author acknowledges the use of open-source algorithmic tools for number verification.